Partial-Label Metric Ceilings for GRN Evaluation in Single-Cell Foundation Model Benchmarks

Abstract

Gene regulatory network (GRN) metrics are often interpreted as direct indicators of biological recovery quality, but curated reference networks are incomplete and context-dependent. We develop a theory-first, low-compute framework for interpreting observed precision/F1/AUPR under partial positive labels and then apply it to existing benchmark outputs in this repository. Under a missing-at-random (MAR) positive-label model, we derive explicit ceilings for observed F1 and AUPR as functions of label coverage `c`, propagate uncertainty in `c`, and stress-test assumption violations with adversarial non-MAR simulations. Across 39 AUPR-evaluable rows (15 methods, 5 references), best ceiling-normalized performance remains limited (best F1 ratio 0.137, best AUPR ratio 0.0138; median AUPR ratio 5.56e-05). Only 3/39 evaluations are above their observed random baseline. The biological interpretation is that low absolute scores are not explained by missing labels alone: after accounting for label incompleteness, substantial model-to-biology mismatch remains, and this mismatch is reference-dependent. We provide a ceiling-aware reporting protocol that reduces both overclaiming and underclaiming in mechanistic GRN studies.

1. Introduction

Foundation-model-based single-cell analyses are increasingly used to propose regulatory interactions [1]. However, evaluating inferred edges against curated resources remains difficult because reference networks are incomplete, biased toward well-studied pathways, and not always matched to tissue context [3,4,7]. In practice, reported AUPR/F1 values conflate at least two effects:

1. **Observation limits** from partial positive labels.
2. **Model headroom** from ranking and calibration quality.

Proposal 1 in this track asks: *what observed metric values are theoretically achievable when labels are incomplete, and how should current results be interpreted relative to those ceilings?*

This paper contributes:

1. A formal metric-ceiling framework for partial labels.
2. An empirical reinterpretation over existing benchmark artifacts.
3. Adversarial tests for coverage misspecification and non-MAR labeling.
4. Biological interpretation of what the results imply for GRN claims in single-cell settings.

2. Biological and Methodological Context

We evaluate predictions generated from single-cell data (Tabula Sapiens context) [2] and compare against references including TRRUST v2 [3], DoRothEA-derived resources [4], and additional benchmark artifacts in this repository. Baseline methods include tree-based and correlation models commonly used in GRN inference [5,6,7].

Two biological realities motivate this study:

1. **Regulation is context-specific.** TF-target interactions are cell-state and tissue dependent [2,4]. A cross-context reference can systematically miss true context-active edges.
2. **Reference curation is biased.** Canonical or well-studied edges are preferentially represented, so observed labels are not a random sample of latent truth [3,4,9].

These realities mean that raw benchmark metrics should not be read as absolute biological validity.

3. Theory

Let:

- U: candidate edge universe.
- T subset U: latent true positives.
- L+ subset T: observed positives.
- c = P(e in L+ | e in T): positive-label coverage.
- P: predicted positive edges.

Define latent and observed metrics:

- Prec_lat = |P intersect T| / |P|, Rec_lat = |P intersect T| / |T|.
- Prec_obs = |P intersect L+| / |P|, Rec_obs = |P intersect L+| / |L+|.

3.1 Proposition 1 (Observed-latent relation under MAR)

If each latent positive is observed independently with probability c (MAR assumption):

- E[Prec_obs] = c * Prec_lat
- E[Rec_obs] = Rec_lat

So latent precision can be approximated by Prec_obs / c when c is estimated.

3.2 Proposition 2 (Observed F1 ceiling)

For an ideal latent predictor (Prec_lat = Rec_lat = 1):

- F1_obs,max(c) = 2c / (1 + c).

3.3 Proposition 3 (Observed AUPR ceiling)

For latent positive count t = |T| under ideal latent ranking:

- AUPR_obs,max(c,t) ~= c + (1-c) * H_t/t where H_t is harmonic. For large t, AUPR_obs,max ~= c.

Observed random baseline under partial labels:

- AUPR_rand,obs = c * b, where b = |T|/|U|.

We report headroom-aware uplift:

- rho_aupr = (AUPR_obs - AUPR_rand,obs) / (AUPR_obs,max - AUPR_rand,obs).

4. Data, Coverage Proxies, and Analysis Design

4.1 Empirical inputs

Primary AUPR/F1 tables:

- network_inference/outputs/score_eval_probe_priors.csv
- network_inference/outputs/score_eval_grn_baselines_immune.csv

Additional F1-only tables:

- network_inference/outputs/score_eval_probe_priors_full_genes.csv
- network_inference/outputs/score_eval_probe_priors_full_genes_crosswalk.csv
- network_inference/outputs/score_eval_probe_priors_full_genes_omnipath.csv

Coverage proxy reports:

- network_inference/outputs/score_eval_probe_priors_missing_report.json
- network_inference/outputs/score_eval_probe_priors_full_genes_missing_report.json
- network_inference/outputs/score_eval_probe_priors_full_genes_crosswalk_missing_report.json
- network_inference/outputs/score_eval_probe_priors_full_genes_omnipath_missing_report.json

4.2 Coverage estimation and uncertainty

Central coverage is overlap-based (`ref_node_overlap_pct/100`) from primary AUPR-capable tables. We model uncertainty with:

- `c ~ Beta(k+1, n-k+1)` where `k=overlap_nodes`, `n=ref_nodes`.

4.3 Adversarial scenarios

Per reference, we evaluate:

- `overlap_point`, `beta_ci_low`, `beta_ci_high`
- misspecification stress: `0.5x`, `1.5x`
- mapping-derived bounds: `mapping_min`, `mapping_max`
- envelope: `conservative_min`, `optimistic_max`

4.4 Synthetic validation

We run:

1. MAR simulations over `c in {0.05,0.1,0.2,0.4,0.7,0.9}` and `t in {50,200,1000,5000}`.
2. Non-MAR simulations where observed labels are rank-biased (optimistic/pessimistic) but keep similar average coverage.

5. Results

5.1 Global reinterpretation: missing labels are not the full story

From `central_aupr_rows.csv`:

- AUPR-evaluable rows: 39
- methods: 15
- references: 5
- best F1 ratio: 0.137393 (`genie3`, `dorothea_trrust_union_immune`)
- best AUPR ratio: 0.013771 (`grnboost2`, `dorothea_trrust_union_immune`)
- median F1 ratio: 0.0
- median AUPR ratio: 5.56e-05

Interpretation: even after accounting for incomplete labels, observed metrics are a small fraction of their theoretical observable headroom. This indicates a **model-limited regime**, not merely a **label-limited regime**.

Evidence: median AUPR ceiling ratio is `5.56e-05` and best ratio is `0.013771`. **Inference:** most methods capture only a tiny fraction of currently observable benchmark signal. **Hypothesis:** current scoring signals are dominated by non-regulatory covariance patterns. **Scientific implication:** benchmark improvement should prioritize biological signal modeling, not only candidate/mapping adjustments.

5.2 Biological interpretation by reference regime

Reference-level medians show two distinct regimes:

1. **Low-coverage transcriptional references** (`dorothea_human`, `dorothea_trrust_union_immune`, `trrust_human`):

- `c` ranges ~0.095 to 0.159.
- AUPR ratios are still low (median ~0.0029 for DoRothEA-based references; ~5.6e-05 for TRRUST).

Biological meaning: low overlap is consistent with context mismatch and curation scope differences across tissues/states [2,4]. But ratios remain low even after coverage correction, so incompleteness alone cannot explain weak recovery.

2. **High-coverage references** (`hpn_dream`, `beeline_gsd`, central `c` ~0.745 and ~0.882):

- Near-zero F1 and tiny AUPR ratios persist.

Biological meaning: when node overlap is high but ranking quality remains near-null, the bottleneck is likely signal-model mismatch (e.g., expression-derived scores vs pathway/causal signaling structure) rather than symbol overlap.

Evidence: high-overlap references (`hpn_dream`, `beeline_gsd`) still show near-zero F1 and very small AUPR ratios. **Inference:** overlap coverage alone does not rescue predictive utility. **Hypothesis:** pathway-level causal structure in these references is not well represented by the evaluated score families. **Scientific implication:** report results by biological reference type and avoid pooled claims across incompatible regulatory layers.

5.3 Most evaluations are below random baseline

`36/39` rows have `AUPR_obs < AUPR_rand,obs` (negative uplift). Only three rows are positive uplift:

- `grnboost2` on `dorothea_trrust_union_immune` (`rho_aupr` = 0.009786)
- `genie3` on `dorothea_trrust_union_immune` (`rho_aupr` = 0.007064)
- `probe_integrated_gradients` on `trrust_human` (`rho_aupr` = 2.27e-05)

Biological meaning: most evaluated edge rankings are not meaningfully enriched for reference edges beyond random expectation in this setup, even after accounting for partial labels.

Evidence: `36/39` rows have negative uplift (`AUPR_obs < c*b`). **Inference:** most reported rankings are below observed-baseline utility in this protocol. **Hypothesis:** thresholding/top-k policies are amplifying noisy edges faster than true regulatory signal. **Scientific implication:** downstream wet-lab prioritization should be conservative unless uplift is positive and robust.

5.4 Method ordering and what it means biologically

Ceiling normalization causes only small central-scenario reordering (one local swap: `probe_perturbation` vs `probe_integrated_gradients`). This indicates that in this dataset the dominant issue is not between-method relative ordering; it is the overall distance of all methods from the estimated biological headroom.

Evidence: only one adjacent rank swap appears under central normalization. **Inference:** relative method ordering is comparatively stable in the central coverage regime. **Hypothesis:** methods share similar error modes on the evaluated references. **Scientific implication:** scientific conclusions should focus on absolute headroom deficit, not marginal leaderboard swaps.

5.5 Adversarial coverage analysis reveals proxy fragility

Overlap/Beta scaling scenarios preserve ranks (Spearman ~1), but mapping-driven scenarios (`mapping_min/max`) produce large shifts (up to 10-11 rank positions on overlapping subsets).

Biological meaning: symbol normalization and curation identity handling can dominate conclusions in sparse-overlap settings. Reported method superiority can be a mapping artifact if coverage assumptions are implicit.

Evidence: mapping-based scenarios shift subset ranks by up to 10-11 positions. **Inference:** coverage proxy choice can be a first-order source of benchmark volatility. **Hypothesis:** alias/identifier ambiguities are concentrated in biologically central genes, disproportionately affecting evaluation. **Scientific implication:** manuscripts must publish mapping policy, ambiguity handling, and sensitivity bands as primary results.

5.6 MAR formulas are accurate under MAR, but non-MAR bias is substantial

MAR validation shows low bias:

- median absolute F1 bias: 7.30e-04
- median absolute AUPR bias: 1.35e-03

Under non-MAR labeling:

- optimistic (high-ranked edges more likely labeled): mean AUPR delta +0.0898
- pessimistic: mean AUPR delta -0.0734

Biological meaning: if curated references overrepresent canonical, easy-to-detect biology, benchmark AUPR can be inflated; if they underrepresent such edges, AUPR can be deflated. This is exactly the curation-bias channel expected in knowledge-driven resources [3,4,9].

Evidence: MAR bias is small, but non-MAR simulations shift mean AUPR by about `+0.0898` (optimistic) or `-0.0734` (pessimistic). **Inference:** curation mechanism, not just coverage magnitude, can change apparent model quality. **Hypothesis:** literature-curated resources over-sample pathway-prominent interactions and under-sample context-specific weak-effect regulation. **Scientific implication:** non-MAR stress tests should be required before making comparative biological claims.

6. Discussion

This study suggests the following biological interpretation of current GRN benchmark behavior:

1. **Low overlap is real but not sufficient as an explanation.** Coverage-corrected ratios remain very low for most settings, so poor scores are not only a gold-standard incompleteness artifact.
2. **Reference choice encodes biological assumptions.** Different references represent different layers of biology (TF-target transcriptional control vs pathway/signal-level interactions). Method performance should be interpreted in that layer-specific context, not pooled uncritically.
3. **Most inferred edges should be treated as exploratory hypotheses.** Given near-random or below-random uplift in most rows, these outputs are currently better suited for triage and follow-up than for strong mechanistic claims.
4. **Curation bias can move conclusions in both directions.** Non-MAR simulations show large AUPR shifts without any change in latent predictor quality. Benchmarks that ignore this are vulnerable to overclaiming and underclaiming.

6.1 Evidence -> Inference -> Hypothesis examples

1. **Evidence:** `36/39` AUPR rows are below the observed random baseline (`AUPR_obs < c*b`), and median normalized uplift is slightly negative. **Inference:** most evaluated ranking functions do not yet enrich true regulatory edges beyond baseline expectation under the current protocol. **Hypothesis:** the dominant signal used by these models is capturing broad co-expression or abundance structure rather than directional regulatory control.
2. **Evidence:** strong performance concentration appears mainly in `dorothea_trrust_union_immune` for tree-based baselines, while other references remain near-null. **Inference:** method success is reference-dependent and likely tied to compatibility between scoring assumptions and reference construction. **Hypothesis:** references emphasizing canonical TF programs reward methods aligned with known high-variance transcriptional modules.
3. **Evidence:** non-MAR perturbations swing mean AUPR by approximately +0.09 (optimistic labeling) or -0.07 (pessimistic labeling). **Inference:** benchmark outcomes are highly sensitive to which true edges are likely to be curated. **Hypothesis:** current curated resources over-sample pathway-prominent or assay-accessible interactions, distorting apparent model quality in both directions.

7. Practical Recommendations for Biological FM Papers

We recommend a mandatory reporting protocol:

1. Report raw metrics and ceiling-normalized metrics together.
2. Report explicit coverage proxy definition and uncertainty interval.
3. Report observed baseline (`c*b`) and uplift (`rho_aupr`).

4. Include at least one non-MAR stress test.
5. Separate claims by biological layer of the reference (transcriptional vs signaling).
6. Treat edges as high-confidence only when they are robust across references/scenarios and show positive uplift.

8. Limitations

- Coverage `c` is proxy-estimated, not identifiable.
- Results are conditioned on existing artifacts and candidate-set definitions.
- Non-MAR simulations are stylized; they show sensitivity, not exact real-world bias magnitude.
- Biological interpretation remains benchmark-centric and does not replace perturbational validation.

9. Conclusion

Partial-label ceilings are necessary for interpreting GRN benchmarks but not sufficient for vindicating weak performance. In this dataset, label incompleteness explains part of the story; substantial model-to-biology mismatch remains after correction. The main value of this framework is not to inflate performance claims, but to make benchmark interpretations honest, uncertainty-aware, and biologically contextualized.

10. Reproducibility and Artifacts

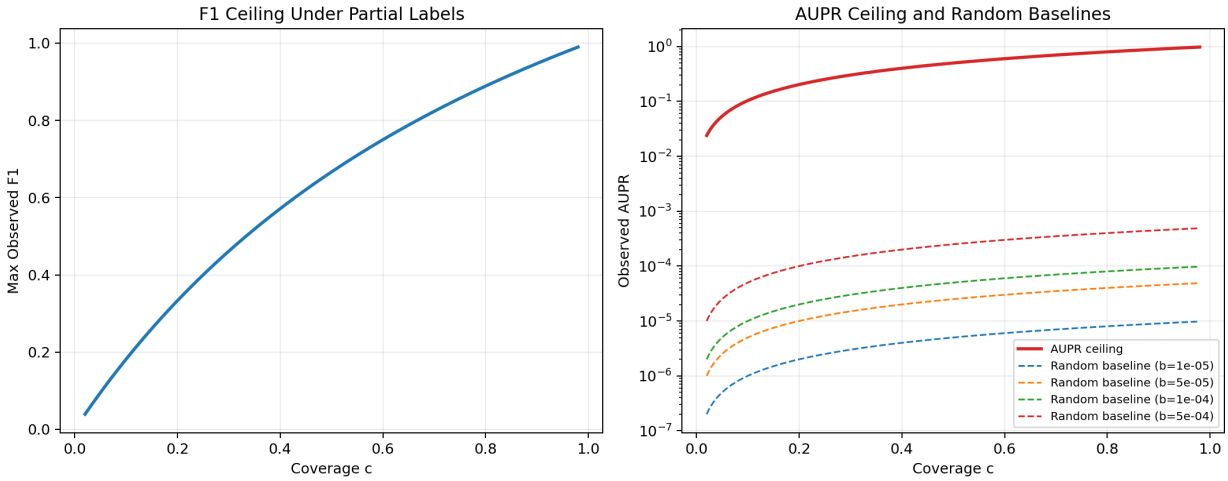
Run:

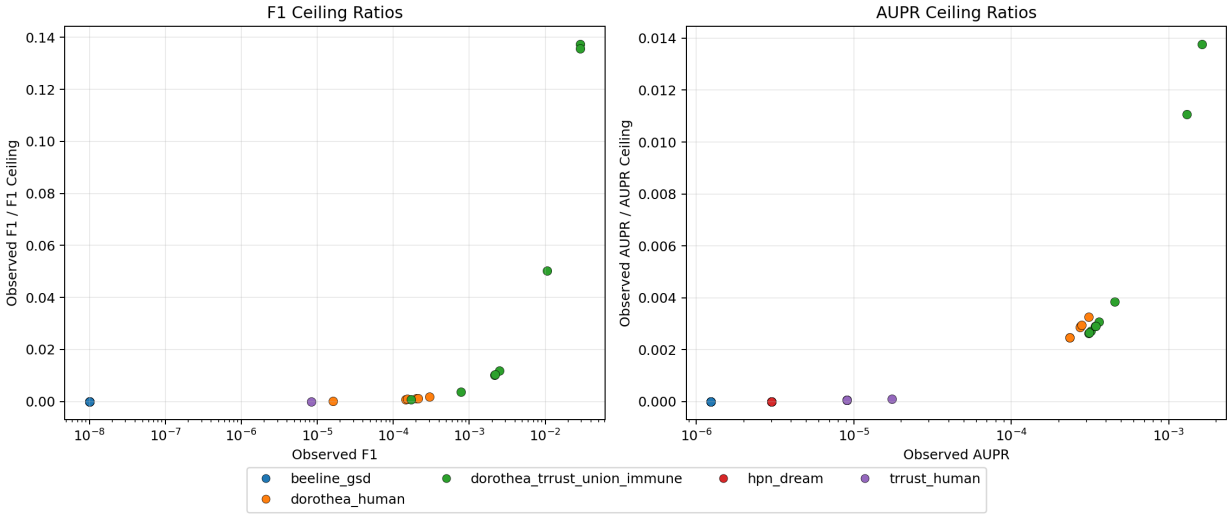
```
python market_research/ambitious_paper_questions/proposal_01_partial_label_metric_ceilings/scripts/run_p
python market_research/ambitious_paper_questions/proposal_01_partial_label_metric_ceilings/scripts/build
```

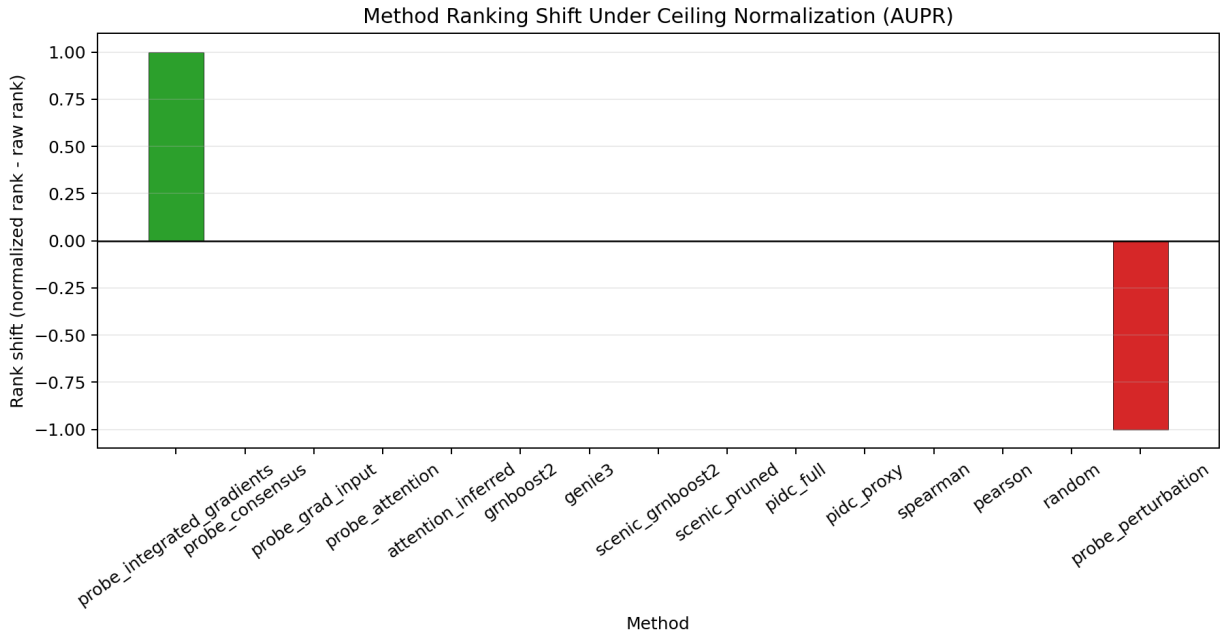
Core outputs:

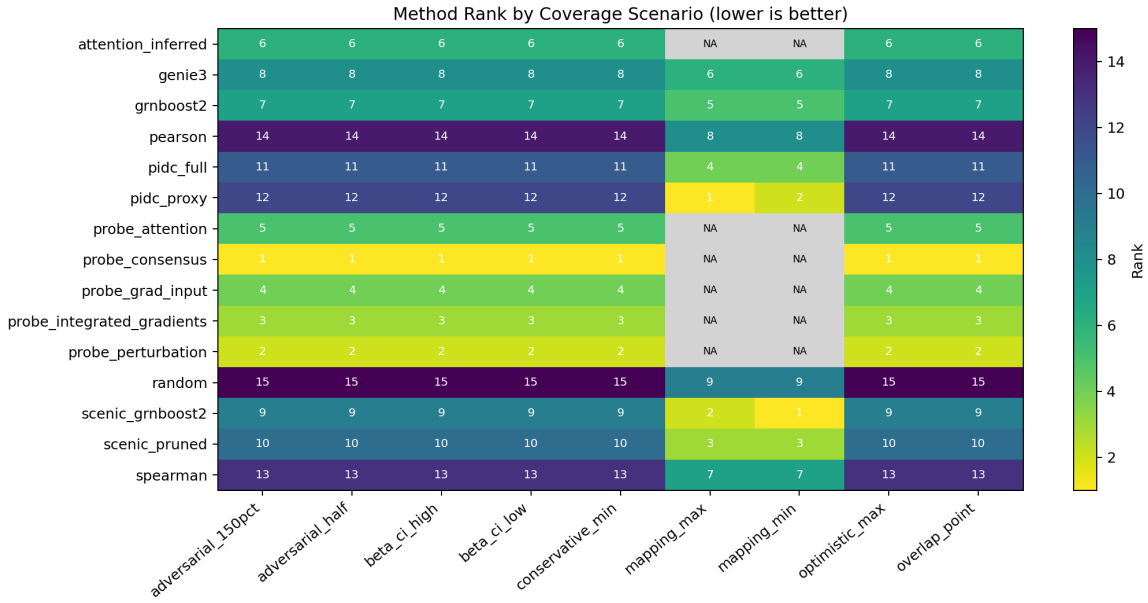
- Tables: `market_research/ambitious_paper_questions/proposal_01_partial_label_metric_ceilings/outputs/`
- Figures: `market_research/ambitious_paper_questions/proposal_01_partial_label_metric_ceilings/outputs`

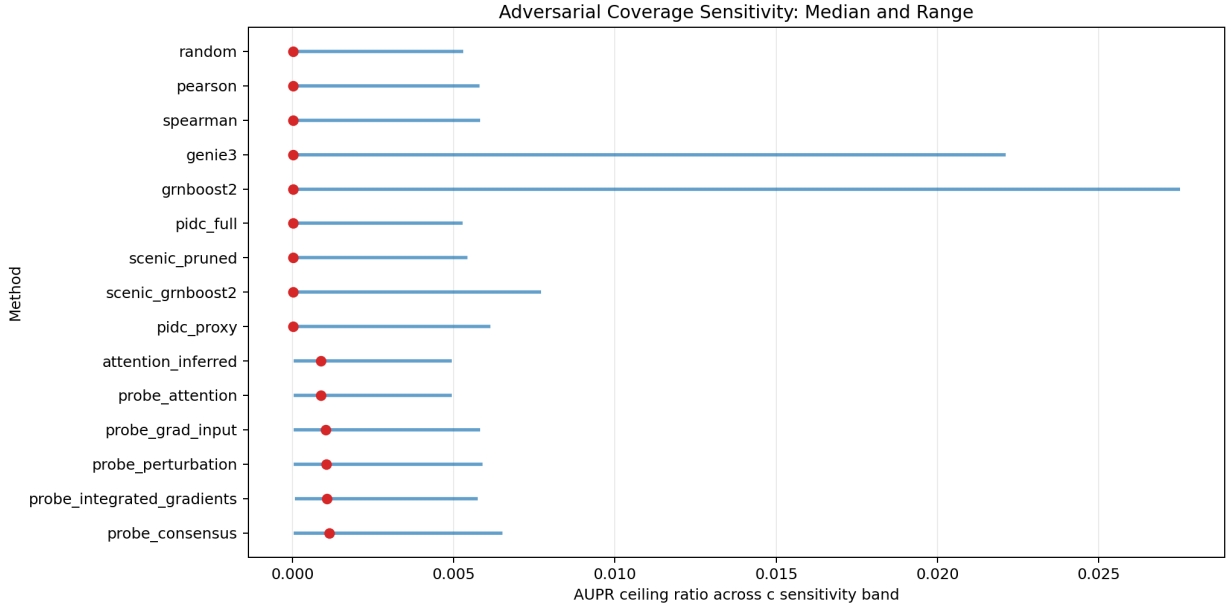
Figures:

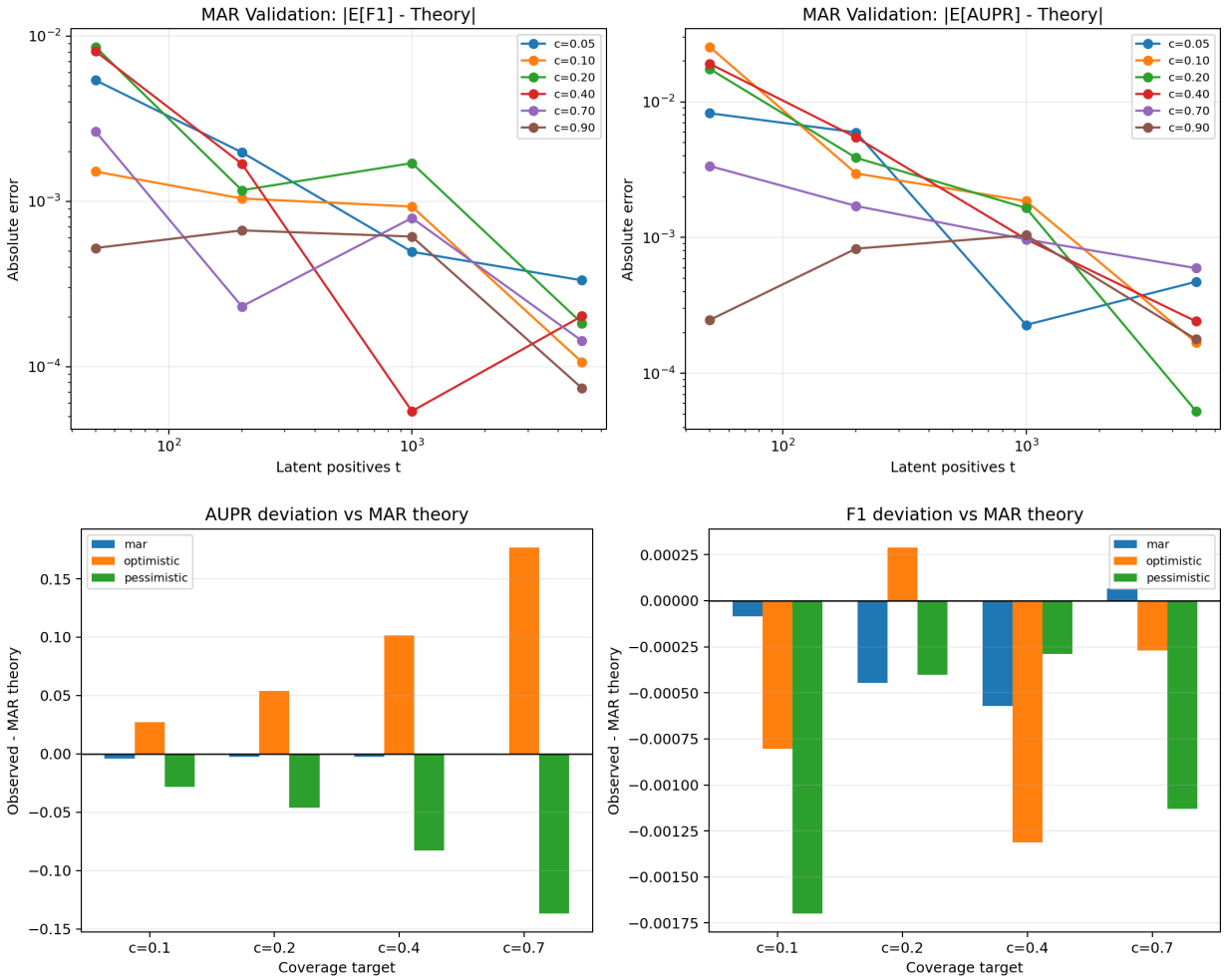



F1 Ceiling Ratios
Observed F1 / F1 Ceiling
Observed F1
AUPR Ceiling Ratios
Observed AUPR / AUPR Ceiling
Observed AUPR
beeline_gsd
dorothea_human
dorothea_trrust_union_immune
hpn_dream
trrust_human


Method Ranking Shift Under Ceiling Normalization (AUPR)
Rank shift (normalized rank - raw rank)
probe_integrated_gradients
probe_consensus
probe_grad_input
probe_attention
attention_inferred
grnboost2
genie3
scenic_grnboost2
scenic_pruned
pidc_full
pidc_proxy
spearman
pearson
random
probe_perturbation
Method


Method Rank by Coverage Scenario (lower is better)
attention_inferred 6 6 6 6 6 NA NA 6 6
genie3 8 8 8 8 8 6 6 8 8
grnboost2 7 7 7 7 7 5 5 7 7
pearson 14 14 14 14 14 8 8 14 14
pidc_full 11 11 11 11 11 4 4 11 11
pidc_proxy 12 12 12 12 12 1 2 12 12
probe_attention 5 5 5 5 5 NA NA 5 5
probe_consensus 1 1 1 1 1 NA NA 1 1
probe_grad_input 4 4 4 4 4 NA NA 4 4
probe_integrated_gradients 3 3 3 3 3 NA NA 3 3
probe_perturbation 2 2 2 2 2 NA NA 2 2
random 15 15 15 15 15 9 9 15 15
scenic_grnboost2 9 9 9 9 9 2 1 9 9
scenic_pruned 10 10 10 10 10 3 3 10 10
spearman 13 13 13 13 13 7 7 13 13
adversarial_150pct
adversarial_half
beta_ci_high
beta_ci_low
conservative_min
mapping_max
mapping_min
optimistic_max
overlap_point
Rank
14
12
10
8
6
4
2


Adversarial Coverage Sensitivity: Median and Range
Method
random
pearson
spearman
genie3
grnboost2
pidc_full
scenic_pruned
scenic_grnboost2
pidc_proxy
attention_inferred
probe_attention
probe_grad_input
probe_perturbation
probe_integrated_gradients
probe_consensus
0.000
0.005
0.010
0.015
0.020
0.025
AUPR ceiling ratio across c sensitivity band



References

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